

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit	MECHANICAL DESIGN CALCULATIONS Columns, drums, desalters, fired-heater coils, nozzles, wind/skirts, weights	CFU-000-ME-CAL-001 Rev A - Issued for review (FEED) 2026-10-02 - Prepared: CLAUDE CODE - Discipline: MECHANICAL
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1. Purpose and scope

FEED-level mechanical design of the CDU/VDU pressure equipment: shell/head/cone thickness to ASME VIII Div.1, external pressure (C-201 full vacuum) with stiffening rings, hydrotest pressures, empty / operating / hydrotest weights, skirt heights, ASCE 7 wind loads with skirt and anchor-bolt checks, API 530 fired-heater tube thickness, and main-column nozzle sizing. Inputs are read from data/equipment.json, streams.json, process_results.json and psv.json (generator cfu/mech). Results are exported to data/mech.json for the layout, civil/foundation and cost-estimate work. Datasheets: CFU-000-ME-DS-001..007; GA drawings: CFU-100/200-ME-GA-001..008.

2. Design basis and assumptions

- Code: ASME BPVC VIII Div.1 (2023) - UG-16, UG-23, UG-27, UG-28, UG-29, UG-32, UG-33, UG-99; ASME II-D Table 1A allowable stresses (Section 3). Flanges ASME B16.5 / B16.47 Group 1.1 ratings.
- Design pressure / temperature and corrosion allowance from equipment.json. Static liquid head to HLL added to the design pressure of bottom courses and bottom heads (operating liquid density from H&MB;).
- Joint efficiency $E = 1.0$ (full RT) for columns, desalters and drums ≥ 2.0 m ID; $E = 0.85$ (spot RT) for smaller drums. Skirt-to-head weld efficiency 0.7.
- Fabrication minimum: $t_{\min}$ (excl. CA) = $\max(\text{UG-16(b)} \ 1.5 \text{ mm}, D/1000 + 2.5 \text{ mm})$; plus CA. Heads: 2:1 semi-ellipsoidal, nominal = $\max(\text{min. after forming} + 1.5 \text{ mm thinning, adjoining shell})$. Cones: 30 deg half-angle, UG-32(g); junction reinforcement per App. 1-5/1-8 by fabricator.
- Cladding / linings (410S, 317L, Monel 400) are corrosion barriers only - not credited for strength; included in weights (3 mm clad, 2 mm Monel lining).
- External pressure (C-201): 1.034 bar (15 psi) full vacuum at design temperature. Factor A from the Div.2 4.4.5.1 closed form (the basis of Fig. G), factor $B = A.E/2$ in the elastic range with a plastic knee at $0.5 \text{ Sy}(T)$ representing chart CS-2. **Stiffening rings: external flat-bar rings at max. 3.0 m spacing** on all C-201 cylindrical courses (rings double as insulation/platform supports); cone-cylinder junctions taken as lines of support (to be confirmed by App. 1-8).
- Wind: ASCE 7-16 Ch.26/29, $V = 150$ mph (67.1 m/s, 3-s gust), Exposure C, $K_d = 0.95$, $K_{zt} = 1.0$, $C_f = 0.7$ (round, moderately smooth), effective diameter = insulated OD + 0.6 m (ladders, piping), platforms $1.2 \text{ m}^2 \times C_f \ 2.0$ each. Gust factor $G = 0.85$ for rigid structures ($n_1 \geq 1 \text{ Hz}$); G_f per 26.11.5 ($\beta = 1 \%$) for flexible columns. Stress checks use ASD $0.6W$ (+ $0.6D$ for anchor uplift).
- Longitudinal stress (UG-23): tension (windward, design pressure, empty weight) $\leq S.E$; compression (leeward, operating weight, + vacuum for C-201) $\leq B$ (UG-23(b), $A = 0.125/(Ro/t)$). No 1.2 wind increase taken.
- Skirt height set by NPSH of the bottoms pump: $LLL \geq \text{pump CL (EL 100.800)} + \text{NPSHr(est.)} + 1.0 \text{ m margin} + 0.6 \text{ m suction losses}$, with minimum 3.0 m (5.0 m C-101, 6.0 m C-201). NPSHr estimated from suction-specific speed 200 (metric) - see datasheet CFU-000-ME-DS-006.
- Weights: steel density 7850 kg/m³; nozzles $1.6 \text{ NPS}^3 \times 1.75 \text{ kg}$ (manways $\times 1.4 + 150 \text{ kg}$); valve trays 60-75 kg/m² + support ring; structured packing 150-210 kg/m³, grid 380 kg/m³; distributors 120 kg/m²; collector trays 220 kg/m²; platforms 250 kg/m² (60 % wrap, 1.2 m wide) + ladders 35 kg/m; insulation mineral wool 130 kg/m³ + 6 kg/m² cladding; skirt fireproofing 50 mm both faces. Operating liquid: tray clear liquid + downcomer backup, packing hold-up 5 %, sump to NLL. Hydrotest: vessel full of water (internals installed, no insulation).
- Accuracy: thickness to nominal plate; weights $\pm 15 \%$ (vessels), $\pm 30 \%$ (fired heaters, exchangers, air coolers). All values to be superseded by vendor/fabricator calculations.

3. Material allowable stresses (ASME II-D Table 1A)

Spec	Material	40 C	100 C	150 C	200 C	250 C	300 C	325 C	350 C	375 C	400 C	425 C	450 C	475 C	500 C
SA-516-70	SA-516 Gr.70 (CS plate, normalised)	138	138	138	138	138	135	132	128	122	101	84	67	51	34
SA-387-11-2	SA-387 Gr.11 Cl.2 (1.25Cr-0.5Mo)	148	148	148	148	148	148	148	148	147	145	142	139	131	107
SA-387-5-2	SA-387 Gr.5 Cl.2 (5Cr-0.5Mo)	148	148	148	148	148	148	148	147	146	144	140	128	99	79

S in MPa; customary-unit values converted and interpolated. SA-516-70 above 425 C is subject to graphitisation limits (not used above 415 C here).

Tag	Base material	Design T (C)	S at design T (MPa)	S at test T (MPa)	E	CA (mm)
C-101	SA-516-70	390	109.3	137.9	1.0	3.0
C-102	SA-516-70	250	137.9	137.9	1.0	3.0
C-103	SA-516-70	330	131.6	137.9	1.0	6.0
C-104	SA-516-70	375	121.6	137.9	1.0	6.0
C-105	SA-516-70	235	137.9	137.9	1.0	3.0
C-106	SA-516-70	185	137.9	137.9	1.0	3.0
C-201	SA-516-70	415	90.8	137.9	1.0	6.0
D-101A	SA-516-70	165	137.9	137.9	1.0	3.0
D-101B	SA-516-70	165	137.9	137.9	1.0	3.0
D-102	SA-516-70	160	137.9	137.9	1.0	6.0
D-103	SA-516-70	90	137.9	137.9	0.85	3.0
D-104	SA-516-70	280	136.4	137.9	1.0	3.0
D-105	SA-516-70	95	137.9	137.9	0.85	3.0
D-106	SA-516-70	120	137.9	137.9	1.0	3.0
D-201	SA-516-70	100	137.9	137.9	0.85	3.0
D-202	SA-516-70	100	137.9	137.9	0.85	3.0

4. Formulae

- Shell, circumferential stress UG-27(c)(1): $t = P.R / (S.E - 0.6 P)$; longitudinal UG-27(c)(2): $t = P.R / (2 S.E + 0.4 P)$; R = inside radius in corroded condition.
- 2:1 ellipsoidal head UG-32(d): $t = P.D / (2 S.E - 0.2 P)$. Cone UG-32(g): $t = P.D / (2 \cos(a) (S.E - 0.6 P))$.
- External pressure UG-28(c): $P_a = 4B / (3 D_o/t)$; UG-33(d) ellipsoidal head: $P_a = B / (R_o/t)$, $R_o = 0.9 D_o$; cone UG-33(f): equivalent cylinder $L_e = (L/2)(1 + D_s/DL)$, $t_e = t \cos(a)$.
- Stiffening ring UG-29: $I_s = D_o^2 L_s (t + A_s/L_s) A / 14$, A from $B = 0.75 P D_o / (t + A_s/L_s)$.
- Hydrotest UG-99(b): $P_t = 1.3 \text{ MAWP } (S_{\text{test}} / S_{\text{design}})$, MAWP taken = design pressure (top); bottom adds water head. Membrane stress at test $\leq 0.9 S_y$ (vertical field test).
- API 530: elastic $t_{\sigma} = P_{el}.D_o / (2 \sigma_{el} + P_{el})$, $t = t_{\sigma} + CA$; rupture $t_{\sigma} = P_{r}.D_o / (2 \sigma_r + P_r)$, $t = t_{\sigma} + f_{\text{corr}}.CA$ ($f_{\text{corr}} = 0.6$).

5. Columns - shell, cone and head thickness (internal pressure)

C-101 - Atmospheric crude fractionator (SA-516-70, P = 3.5 barg, T = 390 C, S = 109.3 MPa, E = 1.0, CA = 3.0 mm)

Component	ID (mm)	z (m from BTL)	P calc (barg)	t circ	t long	t min fab	CA	t req (int)	t ext (FV)	t wind	t nominal (mm)
Stripping section	3,500	0.00-8.67	3.72	5.98	2.98	6.0	3.0	9.0	-	22	22
Swage 30 deg	6,900	8.67-11.61	3.50	12.79	-	9.4	3.0	15.8	-	20	20
Flash zone / rectifying section	6,900	11.61-37.00	3.50	11.08	5.53	9.4	3.0	14.1	-	-	16
Head (top) 2:1 SE	6,900	-	3.50	11.06	-	-	3.0	14.1	-	-	16 (min 14.1 formed)
Head (bottom) 2:1 SE	3,500	-	3.78	6.06	-	-	3.0	9.1	-	-	12 (min 9.1 formed)

C-102 - Kero side stripper (SA-516-70, P = 3.5 barg, T = 250 C, S = 137.9 MPa, E = 1.0, CA = 3.0 mm)

Component	ID (mm)	z (m from BTL)	P calc (barg)	t circ	t long	t min fab	CA	t req (int)	t ext (FV)	t wind	t nominal (mm)
Stripping section	1,700	0.00-9.20	3.65	2.26	1.13	4.2	3.0	7.2	-	-	8
Head (top) 2:1 SE	1,700	-	3.50	2.17	-	-	3.0	7.2	-	-	10 (min 7.2 formed)
Head (bottom) 2:1 SE	1,700	-	3.68	2.27	-	-	3.0	7.2	-	-	10 (min 7.2 formed)

C-103 - Diesel side stripper (SA-516-70, P = 3.5 barg, T = 330 C, S = 131.6 MPa, E = 1.0, CA = 6.0 mm)

Component	ID (mm)	z (m from BTL)	P calc (barg)	t circ	t long	t min fab	CA	t req (int)	t ext (FV)	t wind	t nominal (mm)
Stripping section	1,900	0.00-9.20	3.65	2.65	1.32	4.4	6.0	10.4	-	-	12
Head (top) 2:1 SE	1,900	-	3.50	2.54	-	-	6.0	10.4	-	-	12 (min 10.4 formed)
Head (bottom) 2:1 SE	1,900	-	3.67	2.67	-	-	6.0	10.4	-	-	12 (min 10.4 formed)

C-104 - Ago side stripper (SA-516-70, P = 3.5 barg, T = 375 C, S = 121.6 MPa, E = 1.0, CA = 6.0 mm)

Component	ID (mm)	z (m from BTL)	P calc (barg)	t circ	t long	t min fab	CA	t req (int)	t ext (FV)	t wind	t nominal (mm)
Stripping section	1,400	0.00-9.20	3.66	2.13	1.06	3.9	6.0	9.9	-	-	10
Head (top) 2:1 SE	1,400	-	3.50	2.03	-	-	6.0	9.9	-	-	12 (min 9.9 formed)
Head (bottom) 2:1 SE	1,400	-	3.68	2.14	-	-	6.0	9.9	-	-	12 (min 9.9 formed)

C-105 - Naphtha stabiliser (debutaniser) (SA-516-70, P = 13.0 barg, T = 235 C, S = 137.9 MPa, E = 1.0, CA = 3.0 mm)

Component	ID (mm)	z (m from BTL)	P calc (barg)	t circ	t long	t min fab	CA	t req (int)	t ext (FV)	t wind	t nominal (mm)
Shell	2,100	0.00-18.00	13.13	10.08	5.00	4.6	3.0	13.1	-	-	14
Head (top) 2:1 SE	2,100	-	13.00	9.94	-	-	3.0	12.9	-	-	16 (min 12.9 formed)
Head (bottom) 2:1 SE	2,100	-	13.16	10.06	-	-	3.0	13.1	-	-	16 (min 13.1 formed)

C-106 - Naphtha splitter (SA-516-70, P = 3.5 barg, T = 185 C, S = 137.9 MPa, E = 1.0, CA = 3.0 mm)

Component	ID (mm)	z (m from BTL)	P calc (barg)	t circ	t long	t min fab	CA	t req (int)	t ext (FV)	t wind	t nominal (mm)
Shell	3,100	0.00-29.50	3.65	4.11	2.05	5.6	3.0	8.6	-	12	12
Head (top) 2:1 SE	3,100	-	3.50	3.94	-	-	3.0	8.6	-	-	12 (min 8.6 formed)
Head (bottom) 2:1 SE	3,100	-	3.69	4.16	-	-	3.0	8.6	-	-	12 (min 8.6 formed)

C-201 - Vacuum column (wet, packed) (SA-516-70, P = 3.5 barg + FV, T = 415 C, S = 90.8 MPa, E = 1.0, CA = 6.0 mm)

Component	ID (mm)	z (m from BTL)	P calc (barg)	t circ	t long	t min fab	CA	t req (int)	t ext (FV)	t wind	t nominal (mm)
Boot (stripping / quench)	3,000	0.00-9.33	3.75	6.24	3.11	5.5	6.0	12.2	18	28	28
Lower cone 30 deg	6,600	9.33-12.44	3.50	14.75	-	9.1	6.0	20.7	25	-	25
Main shell (beds 2-4, flash zone)	6,600	12.44-28.59	3.50	12.77	6.37	9.1	6.0	18.8	24	-	25
Upper cone 30 deg	6,600	28.59-30.50	3.50	14.75	-	9.1	6.0	20.7	22	-	22
Top section (bed 1)	4,400	30.50-36.00	3.50	8.52	4.25	6.9	6.0	14.5	21	-	22
Head (top) 2:1 SE	4,400	-	3.50	8.50	-	-	6.0	14.5	19	-	22 (min 19.0 formed)
Head (bottom) 2:1 SE	3,000	-	3.81	6.32	-	-	6.0	12.3	15	-	18 (min 15.0 formed)

6. C-201 external pressure (full vacuum) and stiffening rings

Course	Do (mm)	t nom	t corr/eff	L (mm)	L/Do	Do/t	A	B (MPa)	Pa (bar)	Req. (bar)	t if no rings
Boot (stripping / quench)	3,036	18	12.0	3,000	0.988	253	2.66e-04	23.2	1.222	1.034	25
Lower cone 30 deg	6,650	25	16.5	2,267	0.341	404	4.00e-04	34.6	1.141	1.034	-
Main shell (beds 2-4, flash zone)	6,648	24	18.0	3,000	0.451	369	3.41e-04	29.6	1.070	1.034	43
Upper cone 30 deg	6,644	22	13.9	1,588	0.239	479	4.50e-04	38.8	1.079	1.034	-
Top section (bed 1)	4,442	21	15.0	3,000	0.675	296	3.13e-04	27.2	1.226	1.034	26

Design temperature 415 C, material SA-516-70, CA 6.0 mm; cones use the equivalent-cylinder method. Without rings the main shell would need the thickness in the last column.

Course	No. of rings	Spacing Ls (mm)	Flat bar h x b (mm)	As (mm2)	A	B (MPa)	Is req (10^6 mm4)	I provided (10^6 mm4)
Boot (stripping / quench)	3	3,000	160 x 16	2,560	2.10e-04	18.3	5.33	5.46
Main shell (beds 2-4, flash zone)	5	3,000	300 x 25	7,500	2.88e-04	25.1	55.96	56.25
Top section (bed 1)	1	3,000	220 x 20	4,400	2.40e-04	20.9	16.69	17.75

External rings, continuous fillet welds both sides (UG-30). Ring stiffeners also carry insulation support. Internal bed-support / collector rings are not credited.

Heads under external pressure (UG-33(d)): top head t = 19 mm, bottom head t = 15 mm minimum (corroded 6.0 mm deducted); nominal 22 / 18 mm.

7. Hydrotest

Tag	MAWP (barg)	S_test/S_des	Pt top (barg)	Pt bottom (barg)	sigma bottom (MPa)	0.9 Sy (MPa)	Check	Test position
C-101	3.5	1.262	5.74	9.63	77	234	OK	Vertical (field)
C-102	3.5	1.000	4.55	5.54	59	234	OK	Vertical (field)
C-103	3.5	1.048	4.77	5.76	46	234	OK	Vertical (field)
C-104	3.5	1.134	5.16	6.13	43	234	OK	Vertical (field)
C-105	13.0	1.000	16.90	18.77	142	234	OK	Vertical (field)
C-106	3.5	1.000	4.55	7.60	99	234	OK	Vertical (field)
C-201	3.5	1.518	6.91	10.62	58	234	OK	Vertical (field)
D-101A	17.0	1.000	22.10	22.47	-	-	OK	Horizontal (shop)
D-101B	17.0	1.000	22.10	22.47	-	-	OK	Horizontal (shop)
D-102	3.5	1.000	4.55	4.88	-	-	OK	Horizontal (shop)
D-103	7.0	1.000	9.10	9.19	-	-	OK	Vertical
D-104	3.5	1.011	4.60	4.81	-	-	OK	Horizontal (shop)
D-105	15.2	1.000	19.76	19.93	-	-	OK	Horizontal (shop)
D-106	3.5	1.000	4.55	4.81	-	-	OK	Horizontal (shop)
D-201	3.5	1.000	4.55	4.68	-	-	OK	Horizontal (shop)
D-202	3.5	1.000	4.55	4.61	-	-	OK	Vertical

Foundations for C-101 / C-201 shall be checked for the full-of-water hydrotest weight (Section 8) unless a pneumatic/hydro-pneumatic or horizontal shop test is specified.

8. Weights

Tag	Shell + heads	Skirt + base	Nozzles	Internals	Clad	Platf./ladders	Insul./fireproof.	EMPTY	Op. liquid	OPERATING	Volume (m3)	HYDROTEST
C-101	104.9	14.4	8.8	116.9	13.4	37.7	33.4	329.6	75.4	405.0	1,146	1458.2
C-102	3.7	2.8	1.8	1.4	0.0	4.6	5.7	19.9	2.9	22.9	22	41.2
C-103	6.1	3.4	1.8	1.6	1.4	4.9	7.3	26.5	3.6	30.1	28	53.1
C-104	3.7	2.2	1.7	1.1	1.0	4.3	4.4	18.4	2.1	20.4	15	32.3
C-105	14.7	4.4	2.8	5.3	0.0	9.9	11.0	48.1	5.8	53.9	65	110.7
C-106	29.8	9.6	4.7	24.8	0.0	18.3	21.3	108.7	21.0	129.7	230	334.1
C-201	123.7	11.7	8.8	107.4	15.5	26.4	26.9	320.5	46.4	366.9	822	1128.0

Tonnes. EMPTY = installed, incl. internals, platforms, insulation and fireproofing (erection lift excludes platforms/insulation if installed after setting).

Drums and desalters - thickness and weights

Tag	Service	ID x T/T (m)	Material	P calc (barg)	Shell t req	Shell t nom	Head t req	Head t nom	Fabricated (t)	Empty (t)	Operating (t)	Hydrotest (t)
D-101A	Electrostatic desalter, 1st stage	3.8 x 30.0	SA-516-70	17.29	27.0	28	26.9	28	119.0	137.4	414.0	487.1
D-101B	Electrostatic desalter, 2nd stage	3.8 x 30.0	SA-516-70	17.29	27.0	28	26.9	28	119.0	137.4	414.0	487.1
D-102	Atm. overhead reflux drum (3-phase)	3.4 x 12.0	SA-516-70	3.73	11.9	12	11.9	12	19.4	19.4	63.0	144.0
D-103	Fuel gas knock-out drum	0.9 x 2.5	SA-516-70	7.17	6.4	8	6.4	8	1.2	1.2	1.8	3.0
D-104	Flare knock-out drum (unit)	2.1 x 6.5	SA-516-70	3.64	7.6	8	7.6	8	4.5	5.2	13.9	29.5
D-105	Stabiliser reflux drum	1.7 x 5.5	SA-516-70	15.32	14.2	16	14.2	16	6.2	6.2	11.2	20.4
D-106	Splitter reflux drum	2.7 x 8.5	SA-516-70	3.69	8.2	10	8.2	10	8.9	8.9	27.8	62.8
D-201	Ejector hotwell / sour water separator	1.3 x 5.5	SA-516-70	3.63	6.8	8	6.8	8	2.5	2.5	6.4	10.4
D-202	Vacuum off-gas knock-out drum	0.6 x 2.0	SA-516-70	3.64	6.1	8	6.1	8	0.9	0.9	1.1	1.5

Desalters operate liquid-full (crude/water, 780 kg/m³); include 3 transformers (4.5 t each) and 18 t internals. Drums at 50 % liquid. Boots included (D-102, D-105).

9. Skirt height, wind load and longitudinal stress

Tag	Bottoms pump	NPSHr est. (m)	LLL above BTL (m)	Skirt height (m)	BTL elevation	Skirt t (mm)	Skirt OD (mm)
C-101	P-112A/B	4.2	0.5	6.5	106.500	20	3,544
C-102	P-109A/B	1.9	0.5	4.0	104.000	8	1,716
C-103	P-110A/B	2.2	0.5	4.5	104.500	8	1,924
C-104	P-111A/B	1.5	0.5	3.5	103.500	8	1,420
C-105	- (no pump)	-	0.5	6.0	106.000	8	2,128
C-106	P-117A/B	5.4	0.5	7.5	107.500	12	3,124
C-201	P-204A/B	2.6	0.6	6.0	106.000	20	3,056

C-105 bottoms flow under pressure to C-106 / kettle E-116; skirt 6.0 m for kettle liquid head and bottom piping.

Tag	Height (m)	n1 (Hz)	Type	G / Gf	Kz top	qh (kPa)	Base shear (kN)	Base moment (kNm)	ASD moment (kNm)	Vortex Vcr (m/s)
C-101	45.2	0.77	flexible	1.156	1.375	3.60	801	21,781	13,069	27.5
C-102	13.6	4.25	rigid	0.890	1.068	2.80	60	485	291	39.3
C-103	14.2	4.26	rigid	0.889	1.077	2.82	67	562	337	44.8
C-104	13.0	3.59	rigid	0.890	1.059	2.77	53	416	250	28.7
C-105	24.5	1.66	rigid	0.882	1.209	3.17	138	1,992	1,195	18.6
C-106	37.8	1.16	rigid	0.876	1.324	3.47	294	6,308	3,785	18.9
C-201	43.1	0.74	flexible	1.225	1.361	3.57	692	17,661	10,597	17.1

Strength-level wind. Vortex shedding: critical velocity $V_{cr} = n_1 D / 0.2$. Columns with $V_{cr} < \sim 25$ m/s and $D/t > 200$ to be checked in detail (helical strakes not expected for insulated columns with platforms).

C-101 - wind / weight longitudinal stress check (corroded, ASD)

Section (bottom of)	Elev. above grade (m)	t nom	M ASD (MNm)	sig P	sig M	sig W	Tension	Allow. S.E	Compression	Allow. B	Result
Stripping section	6.50	22	10	16.1	54.3	17.4	56.5	109.3	71.7	80.1	OK
Swage 30 deg	15.17	20	6	20.8	44.4	19.4	49.2	109.3	63.7	74.1	OK
Flash zone / rectifying section	18.11	16	5	46.4	10.7	10.6	48.4	109.3	21.3	40.9	OK
Skirt (base, 1.5 mm CA, E = 0.7)	0.00	20	13	-	70.9	18.4	56.0	96.5	89.3	100.9	OK

Anchor bolts C-101: 28 x M64 ASTM F1554 Gr.55 on 3.79 m BCD; max. tension 430 kN/bolt (0.6D + 0.6W, empty), required root area 2262 mm². Hydrotest skirt compression 86.8 MPa (25 % wind). Base shear 801 kN, base moment 21781 kNm (strength) for foundation design.

C-201 - wind / weight longitudinal stress check (corroded, ASD)

Section (bottom of)	Elev. above grade (m)	t nom	M ASD (MNm)	sig P	sig M	sig W	Tension	Allow. S.E	Compression	Allow. B	Result
Boot (stripping / quench)	6.00	28	8	11.9	51.8	16.0	49.9	90.8	71.3	81.3	OK
Lower cone 30 deg	15.33	25	5	16.0	40.4	18.7	39.7	90.8	63.8	78.4	OK
Main shell (beds 2-4, flash zone)	18.44	25	4	30.4	5.7	6.9	29.9	90.8	21.6	58.1	OK
Upper cone 30 deg	34.59	22	0	41.7	0.9	1.9	40.8	90.8	15.1	44.5	OK
Top section (bed 1)	36.50	22	0	24.1	1.1	2.0	23.4	90.8	10.2	67.7	OK
Skirt (base, 1.5 mm CA, E = 0.7)	0.00	20	11	-	77.2	19.4	60.3	96.5	96.6	105.6	OK

Anchor bolts C-201: 24 x M64 ASTM F1554 Gr.55 on 3.31 m BCD; max. tension 462 kN/bolt (0.6D + 0.6W, empty), required root area 2433 mm². Hydrotest skirt compression 81.2 MPa (25 % wind). Base shear 692 kN, base moment 17661 kNm (strength) for foundation design.

10. Fired heater tubes - API 530 (9Cr-1Mo, 100,000 h)

Item	H-101 radiant	H-201 radiant
Radiant average flux (kW/m ² , OD)	31.5	25.0
Peak flux (x1.8 circ. x1.1 long.) (kW/m ²)	62.4	49.5
Inside peak flux (kW/m ²)	71.7	56.9
Bulk fluid T at outlet (C)	365.7	398.4
Inside film coefficient (W/m ² K, EOR)	1,100	650
Coke/fouling resistance EOR (m ² K/W)	0.0005	0.0009
dT film (C)	65.2	87.6
dT coke (C)	35.9	51.2
dT wall (C)	27.2	21.6
Estimated max. TMT EOR (C)	494	559
Design metal temperature (TMT + 15 C)	510	575
equipment.json des_T (C)	545	580
Elastic design pressure (MPa)	3.85	1.98
Rupture design pressure (MPa, coil inlet op.)	1.70	1.90
sigma_el at Tdm (MPa)	101.8	78.0
sigma_r 100,000 h at Tdm (MPa)	83.0	35.0
Corrosion allowance (mm)	3.0	3.0
Elastic t_sigma (mm)	3.12	2.11
Rupture t_sigma (mm)	1.71	4.45
Elastic t_min = t_sigma + CA	6.12	5.11
Rupture t_min = t_sigma + f_corr CA	3.51	6.25
Minimum thickness (mm)	6.12	6.25
Governing	elastic	rupture
Minimum schedule	Sch 40 (7.11 mm)	Sch 40 (7.11 mm)
Selected (OD 168.3 mm, A335 P9)	Sch 80 (10.97 mm)	Sch 80 (10.97 mm)

Sch 80 is retained (process mass-flux basis ID 146.3 mm) - gives margin for coke spalling/erosion and decoking. 9Cr-1Mo stresses digitised from API 530 curves: 425 C: 123/205, 450 C: 118/160, 475 C: 112/123, 500 C: 105/93, 525 C: 97/68, 550 C: 88/49, 575 C: 78/35, 600 C: 67/25 MPa (sigma_el/sigma_r).

Heater	Fired (MW)	APH duty (MW)	Radiant (MW)	Flue gas (t/h)	BWT (C)	Crossover (C)	Flue out conv. (C)	LMTD (C)	Conv. bare area (m2)	Rows (shock+studded+coils)	Utility coil req. (MW)
H-101	67.4	6.4	41.7	96	954	302	381	302	563	2 + 6 + 2 SS + 0 util.	0.00
H-201	17.5	0.0	10.3	25	822	364	399	184	188	2 + 3 + 0 SS + 2 util.	0.77

Convection estimate: $U = 105 \text{ W/m}^2\text{K}$ on bare-tube area (studded), 50 C minimum approach to process inlet. Vendor to optimise.

11. Nozzle sizing summary (main columns)

C-101 - Atmospheric crude fractionator

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N1	Feed - transfer line from H-101 (tangential, vapour horn)	36"	150#	12.26	34,016	17	15.0	two-phase feed $\rho_{m.v2} \leq 6000 \text{ Pa}$
N2	Overhead vapour to A-101	36"	150#	37.00	45,065	5.09	19.9	vapour outlet $\rho_{m.v2} \leq 3000 \text{ Pa}$
N3	Reflux from P-103	8"	150#	35.65	145	698	1.2	reflux / pumped liquid $\leq 2.0 \text{ m/s}$
N4	TPA return	8"	150#	35.60	246	679	2.1	pumped liquid return $\leq 2.5 \text{ m/s}$
N5	TPA draw to P-106 (draw sump)	14"	150#	33.63	270	617	0.8	gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting)
N8	MPA return	10"	150#	29.05	320	650	1.7	pumped liquid return $\leq 2.5 \text{ m/s}$
N9	MPA draw to P-107 (draw sump)	16"	150#	27.08	352	591	0.8	gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting)
N12	BPA return	8"	150#	21.28	263	630	2.3	pumped liquid return $\leq 2.5 \text{ m/s}$
N13	BPA draw to P-108 (draw sump)	16"	150#	19.31	289	573	0.7	gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting)
N6	Kero draw to C-102	12"	150#	29.21	128	591	0.5	gravity draw to side stripper $\leq 0.6 \text{ m/s}$
N7	Vapour return from C-102	8"	150#	29.72	2,250	3.70	19.4	vapour return $\rho_{m.v2} \leq 4000 \text{ Pa}$
N10	Diesel draw to C-103	12"	150#	21.44	157	573	0.6	gravity draw to side stripper $\leq 0.6 \text{ m/s}$
N11	Vapour return from C-103	8"	150#	21.95	2,621	3.75	22.6	vapour return $\rho_{m.v2} \leq 4000 \text{ Pa}$
N14	Ago draw to C-104	10"	150#	14.89	79	636	0.4	gravity draw to side stripper $\leq 0.6 \text{ m/s}$
N15	Vapour return from C-104	6"	150#	15.40	1,382	3.97	20.6	vapour return $\rho_{m.v2} \leq 4000 \text{ Pa}$
N16	Stripping steam (from H-101 SS coil)	10"	150#	4.72	4,886	1.56	26.7	stripping steam $\leq 35 \text{ m/s}$
N17	Atm. residue to P-112	16"	150#	-0.88	365	704	0.9	pump suction (bottoms) $\leq 1.0 \text{ m/s}$
N18	Vent / steam-out	3"	150#	37.00	-	-	-	-
N19	PSV-1001 relief (4 x T orifice)	4 x 8"	150#	37.00	-	-	-	-
N20A/B	LT / LG bridle (bottom sump)	2 x 3"	150#	0.20	-	-	-	-
N21	Wash-zone overflash / slop connection (spare)	4"	150#	13.06	-	-	-	-
M1	Manway 24" (davit)	24"	150#	36.30	-	-	-	-
M2	Manway 24" (davit)	24"	150#	30.27	-	-	-	-
M3	Manway 24" (davit)	24"	150#	24.34	-	-	-	-
M4	Manway 24" (davit)	24"	150#	17.18	-	-	-	-
M5	Manway 24" (davit)	24"	150#	12.11	-	-	-	-
M6	Manway 24" (davit)	24"	150#	6.84	-	-	-	-
M7	Manway 24" (davit)	24"	150#	4.07	-	-	-	-

C-201 - Vacuum column (wet, packed)

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N1	Feed - transfer line from H-201 (tangential to vapour horn)	66"	150#	14.14	464,525	0.56	59.8	vacuum transfer line $\leq 70 \text{ m/s}$ ($\rho_{m.v2} \ll 6000 \text{ Pa}$)
N2	Overhead vapour to J-201 / E-202	48"	150#	36.00	229,101	0.01	56.3	vacuum vapour $\leq 60 \text{ m/s}$
N3	LVGO PA return (+ LVGO reflux)	6"	150#	35.30	96	860	1.4	pumped liquid return $\leq 2.5 \text{ m/s}$
N4	LVGO total draw to P-201	12"	150#	31.25	226	739	0.9	gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting)

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N5	LVGO internal reflux to bed 2	4"	150#	28.45	45	760	1.5	pumped liquid return <= 2.5 m/s
N6	HVGO total draw to P-202	16"	150#	23.64	374	708	0.9	gravity draw / pump suction <= 0.9 m/s (self-venting)
N7	HVGO PA return	8"	150#	23.34	204	790	1.8	pumped liquid return <= 2.5 m/s
N8	Wash oil (HVGO) to bed 4	2"	150#	18.45	16	790	2.1	pumped liquid return <= 2.5 m/s
N9	Slop wax draw to P-203	3"	150#	15.60	11	714	0.6	gravity draw / pump suction <= 0.9 m/s (self-venting)
N10	Stripping steam	6"	150#	6.30	1,498	1.86	22.3	stripping steam <= 35 m/s
N11	VR quench return (from E-201 outlet)	3"	150#	3.75	30	905	1.7	pumped liquid return <= 2.5 m/s
N12	Vacuum residue to P-204	10"	150#	-0.75	179	757	1.0	pump suction (bottoms) <= 1.0 m/s
N13	PSV-2001 relief (R)	6"	150#	36.00	-	-	-	-
N14A/B	LT / LG bridle (boot)	2 x 3"	150#	0.30	-	-	-	-
N15	Vent / steam-out / N2 purge	4"	150#	36.00	-	-	-	-
M1	Manway 24"	24"	150#	30.95	-	-	-	-
M2	Manway 24"	24"	150#	24.84	-	-	-	-
M3	Manway 24"	24"	150#	18.75	-	-	-	-
M4	Manway 24" (36" in flash zone)	36"	150#	13.24	-	-	-	-
M5	Manway 24"	24"	150#	4.40	-	-	-	-

C-102 - Kero side stripper

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N1	Kero feed from C-101 tray 10	12"	150#	8.00	128	591	0.5	gravity draw to side stripper <= 0.6 m/s
N2	Vapour return to C-101	8"	150#	9.20	2,250	3.70	19.4	vapour return rho.v2 <= 4000 Pa
N3	Stripping steam	4"	150#	4.05	944	1.56	31.9	stripping steam <= 35 m/s
N4	Bottoms to P-109	8"	150#	-0.42	114	603	1.0	pump suction (bottoms) <= 1.0 m/s
N5A/B	LT / LG bridle	2 x 2"	150#	0.20	-	-	-	-
N6	Vent / steam-out / PSV-1009 (common)	3"	150#	9.20	-	-	-	-
M1	Manway 24"	24"	150#	8.45	-	-	-	-
M2	Manway 24"	24"	150#	3.30	-	-	-	-

C-103 - Diesel side stripper

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N1	Diesel feed from C-101 tray 22	12"	150#	8.00	157	573	0.6	gravity draw to side stripper <= 0.6 m/s
N2	Vapour return to C-101	8"	150#	9.20	2,621	3.75	22.6	vapour return rho.v2 <= 4000 Pa
N3	Stripping steam	6"	150#	4.05	1,066	1.56	15.9	stripping steam <= 35 m/s
N4	Bottoms to P-110	10"	150#	-0.47	140	584	0.8	pump suction (bottoms) <= 1.0 m/s
N5A/B	LT / LG bridle	2 x 2"	150#	0.20	-	-	-	-
N6	Vent / steam-out / PSV-1009 (common)	3"	150#	9.20	-	-	-	-
M1	Manway 24"	24"	150#	8.45	-	-	-	-
M2	Manway 24"	24"	150#	3.30	-	-	-	-

C-104 - Ago side stripper

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N1	Ago feed from C-101 tray 32	10"	150#	8.00	79	636	0.4	gravity draw to side stripper <= 0.6 m/s
N2	Vapour return to C-101	6"	150#	9.20	1,382	3.97	20.6	vapour return rho.v2 <= 4000 Pa
N3	Stripping steam	3"	150#	4.05	575	1.56	33.5	stripping steam <= 35 m/s
N4	Bottoms to P-111	8"	150#	-0.35	71	649	0.6	pump suction (bottoms) <= 1.0 m/s

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N5A/B	LT / LG bridle	2 x 2"	150#	0.20	-	-	-	-
N6	Vent / steam-out / PSV-1009 (common)	3"	150#	9.20	-	-	-	-
M1	Manway 24"	24"	150#	8.45	-	-	-	-
M2	Manway 24"	24"	150#	3.30	-	-	-	-

C-105 - Naphtha stabiliser (debutaniser)

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N1	Feed (tray 13)	8"	300#	8.98	187	614	1.6	pumped liquid return <= 2.5 m/s
N2	Overhead vapour to A-106	8"	300#	18.00	856	22	7.4	vapour outlet rho.v2 <= 3000 Pa
N3	Reflux	3"	300#	16.35	23	530	1.3	reflux / pumped liquid <= 2.0 m/s
N4	Reboiler feed to E-116 (kettle)	18"	300#	-0.53	431	554	0.8	gravity draw / pump suction <= 0.9 m/s (self-venting)
N5	Reboiler return from E-116 (kettle)	10"	300#	4.22	614	212	3.4	reboiler vapour return rho.v2 <= 4000 Pa
N6	Stabilised naphtha (bottoms) to E-114/C-106	12"	300#	0.30	196	554	0.7	pump suction (bottoms) <= 1.0 m/s
N7	PSV-1005 relief (R)	6"	300#	18.00	-	-	-	-
N8A/B	LT / LG bridle	2 x 2"	300#	0.20	-	-	-	-
N9	Vent / steam-out	2"	300#	18.00	-	-	-	-
M1	Manway 24"	24"	300#	17.00	-	-	-	-
M2	Manway 24"	24"	300#	8.98	-	-	-	-
M3	Manway 24"	24"	300#	3.52	-	-	-	-

C-106 - Naphtha splitter

Mark	Service	Size	Rating	z from BTL (m)	Flow (m3/h act.)	rho (kg/m3)	v (m/s)	Criterion
N1	Feed (tray 21)	8"	150#	15.60	196	554	1.7	pumped liquid return <= 2.5 m/s
N2	Overhead vapour to A-107	20"	150#	29.50	14,876	5.72	22.0	vapour outlet rho.v2 <= 3000 Pa
N3	Reflux	6"	150#	27.85	90	640	1.3	reflux / pumped liquid <= 2.0 m/s
N4	Reboiler feed to E-117 (thermosyphon)	16"	150#	-0.78	370	624	0.9	gravity draw / pump suction <= 0.9 m/s (self-venting)
N5	Reboiler return from E-117 (thermosyphon)	20"	150#	4.13	11,154	6.89	21.5	reboiler vapour return rho.v2 <= 4000 Pa
N6	Bottoms to P-117A/B	10"	150#	0.30	130	624	0.7	pump suction (bottoms) <= 1.0 m/s
N7	PSV-1007 relief (T)	8"	150#	29.50	-	-	-	-
N8A/B	LT / LG bridle	2 x 2"	150#	0.20	-	-	-	-
N9	Vent / steam-out	2"	150#	29.50	-	-	-	-
M1	Manway 24"	24"	150#	28.50	-	-	-	-
M2	Manway 24"	24"	150#	21.70	-	-	-	-
M3	Manway 24"	24"	150#	15.61	-	-	-	-
M4	Manway 24"	24"	150#	9.51	-	-	-	-
M5	Manway 24"	24"	150#	3.43	-	-	-	-

12. Shell & tube exchangers and air coolers - mechanical summary

Tag	TEMA size	Shells	m2/shell	Tubes/shell	Tube OD/L (mm)	Des. P sh/tu (barg)	Des. T	Shell matl.	Shell t	Channel t	Empty t/shell	Hydro t/shell
E-101	1200-6096 AES	1	359	756	25.4 / 6096	30 / 20	185	SA-516-70	20	14	18.9	27.2
E-102	1300-6096 AES	1	438	923	25.4 / 6096	30 / 20	240	SA-516-70	20	16	22.4	32.0
E-103	1250-6096 AES	2	402	848	25.4 / 6096	30 / 20	255	SA-516-70	20	14	20.7	29.6
E-104	1200-6096 AES	1	378	797	25.4 / 6096	30 / 20	235	SA-516-70	20	14	19.5	27.7
E-105	1300-6096 AES	1	431	910	25.4 / 6096	30 / 20	285	SA-516-70	22	16	22.7	32.3

Tag	TEMA size	Shells	m2/shell	Tubes/shell	Tube OD/L (mm)	Des. P sh/tu (barg)	Des. T	Shell matl.	Shell t	Channel t	Empty t/shell	Hydro t/shell
E-106	1350-6096 AES	2	500	1055	25.4 / 6096	30 / 20	270	SA-516-70	22	16	25.2	35.6
E-107	1150-6096 AES	1	324	683	25.4 / 6096	30 / 20	325	SA-516-70	22	18	18.2	25.8
E-108	1400-6096 AES	4	549	1157	25.4 / 6096	30 / 20	345	SA-387-11-2	25	18	28.2	39.3
E-109	1000-6096 AES	1	263	555	25.4 / 6096	30 / 20	365	SA-387-11-2	20	16	14.5	20.2
E-110	1300-6096 AES	2	443	934	25.4 / 6096	30 / 20	345	SA-387-11-2	22	18	23.2	32.9
E-111	1400-6096 AES	2	547	1154	25.4 / 6096	30 / 20	395	SA-387-11-2	25	18	28.1	39.3
E-113	1300-6096 AKT	1	223	627	19.05 / 6096	13 / 20	290	SA-516-70	14	12	10.3	20.2
E-201	900-6096 AKT	1	84	236	19.05 / 6096	6 / 20	310	SA-516-70	12	10	4.9	9.6
E-114	650-6096 AES	1	86	182	25.4 / 6096	20 / 20	235	SA-516-70	10	10	5.0	7.4
E-115	900-6096 AEU	2	370	1040	19.05 / 6096	3.5 / 7	90	SA-516-70	12	12	9.0	13.5
E-116	1300-6096 BKT	1	242	680	19.05 / 6096	15 / 45	430	SA-516-70	18	32	12.9	22.8
E-117	750-6096 BXM	1	225	632	19.05 / 6096	5 / 13	280	SA-516-70	10	10	8.0	11.2
E-118	600-6096 AEL	1	84	177	25.4 / 6096	20 / 20	160	SA-516-70	10	10	4.7	6.8
E-202	1100-6096 AXS	1	549	1543	19.05 / 6096	3.5 / 7 (FV)	100	SA-516-70	14	14	18.6	25.4
E-203	500-6096 AXS	1	79	222	19.05 / 6096	3.5 / 7 (FV)	100	SA-516-70	10	10	3.5	5.0
E-204	400-4877 AXS	1	30	107	19.05 / 4877	3.5 / 7 (FV)	100	SA-516-70	10	10	1.9	2.7

Shell t = max(UG-27 with E = 0.85 + CA, TEMA R minimum). Bundle diameter from tube-count correlation (square pitch 1.25 OD for fouling crude; triangular for clean condensers).

Tag	Service	Bare area (m2)	Bays (data)	Rows req.	Bays @ 6 rows	Fans	Fan D (m)	Motor kW	Empty (t)
A-101	Atm. overhead condenser	1,667	3	7	4	6	4.5	55	118.7
A-103	Kero cooler	184	1	3	1	2	4.5	11	23.8
A-104	Diesel cooler	302	1	4	1	2	4.5	18.5	28.8
A-105	Ago cooler	287	1	4	1	2	4.5	30	28.2
A-106	Stabiliser overhead condenser	214	1	3	1	2	4.5	11	25.1
A-107	Splitter overhead condenser	881	2	6	2	4	4.5	30	69.4
A-108	Heavy naphtha product cooler	401	1	5	1	2	4.5	30	33.0
A-201	Lvgo cooler	397	1	5	1	2	4.5	15	32.8
A-202	Hvgo Product cooler	361	1	5	1	2	4.5	37	31.3

13. Process-data issues and holds

#	Item	Issue / recommendation
1	H-101	Radiant fraction 0.65 (basis.DESIGN) gives an estimated bridgewall temperature of 954 C with APH; API 560 practice <= ~900-950 C. Suggest radiant fraction ~0.70 (more radiant surface) - vendor to optimise.
2	H-201	88 % efficiency is not achievable with process convection alone: charge enters at 349 C, so flue gas cannot be cooled below ~399 C (max. process convection 3.6 MW vs 4.4 MW assumed). ~0.8 MW must go to a utility coil (steam superheat / BFW) or an APH - none exists in equipment.json. Also vacuum-heater outlet tubes normally step up in size (6"->8"->10") - all tubes are 6" in the data.
3	P-206A/B	Rated flow 0.65 m3/h at 60 m is outside the centrifugal (API 610) range - use a positive-displacement / metering pump (API 674/675).
4	P-118	Single pump without spare (desalter mud-wash/recycle) - confirm intermittent duty.
5	PSV-1001	4 x T orifices for 'Reflux failure / blocked OH (total OH vapour)' (216 t/h, set 2.4 barg vs MAWP 3.5 barg). Raising set to MAWP and crediting unaffected pumparound duty (API 521 4.4.3) should reduce this.
6	PSV-1009	One PSV protects C-102/103/104: requires no isolation between the vessels (locked-open valves) - confirm in P&ID.;
7	E-1xx/2xx	Shell IDs from tube-count/bundle correlation exceed equipment.json 'D' for: E-113 1.30 vs 0.9, E-116 1.30 vs 0.9 m (kettles include the enlarged shell).
8	Design P	Only C-201 (and ejector condensers) carry FV. Columns C-101..C-106 and drums D-101A, D-101B, D-102, D-103, D-104, D-106, D-201 are not full-vacuum capable as designed; refinery practice is FV for steam-out (or written steam-out procedure + vacuum breakers). Process to confirm.

14. Summary of results

Tag	Service	Size (m)	Material	Shell t (mm)	Head t (mm)	Skirt (m / mm)	Empty (t)	Operating (t)	Hydrotest (t)
C-101	Atmospheric crude fractionator	6.9 / 3.5 x 37	SA-516-70	16-22	16/12	6.5 / 20	330	405	1,458

Tag	Service	Size (m)	Material	Shell t (mm)	Head t (mm)	Skirt (m / mm)	Empty (t)	Operating (t)	Hydrotest (t)
C-102	Kero side stripper	1.7 x 9.2	SA-516-70	8	10/10	4 / 8	20	23	41
C-103	Diesel side stripper	1.9 x 9.2	SA-516-70	12	12/12	4.5 / 8	27	30	53
C-104	Ago side stripper	1.4 x 9.2	SA-516-70	10	12/12	3.5 / 8	18	20	32
C-105	Naphtha stabiliser (debutaniser)	2.1 x 18	SA-516-70	14	16/16	6 / 8	48	54	111
C-106	Naphtha splitter	3.1 x 29.5	SA-516-70	12	12/12	7.5 / 12	109	130	334
C-201	Vacuum column (wet, packed)	6.6 / 3 / 4.4 x 36	SA-516-70	22-28	22/18	6 / 20	320	367	1,128
D-101A	Electrostatic desalter, 1st stage	3.8 x 30	SA-516-70	28	28	saddles	137	414	487
D-101B	Electrostatic desalter, 2nd stage	3.8 x 30	SA-516-70	28	28	saddles	137	414	487
D-102	Atm. overhead reflux drum (3-phase)	3.4 x 12	SA-516-70	12	12	saddles	19	63	144
D-103	Fuel gas knock-out drum	0.9 x 2.5	SA-516-70	8	8	legs	1	2	3
D-104	Flare knock-out drum (unit)	2.1 x 6.5	SA-516-70	8	8	saddles	5	14	29
D-105	Stabiliser reflux drum	1.7 x 5.5	SA-516-70	16	16	saddles	6	11	20
D-106	Splitter reflux drum	2.7 x 8.5	SA-516-70	10	10	saddles	9	28	63
D-201	Ejector hotwell / sour water separator	1.3 x 5.5	SA-516-70	8	8	saddles	3	6	10
D-202	Vacuum off-gas knock-out drum	0.6 x 2	SA-516-70	8	8	legs	1	1	1
H-101	Atmospheric crude charge heater	21.9 x 12.7 x 38	A335 P9 tubes	10.97 (tube)	-	-	1,077	1,106	954
H-201	Vacuum heater	21.9 x 5.4 x 32	A335 P9 tubes	10.97 (tube)	-	-	340	349	360